Lightning Symposium 2024

Year One Symposium: Progress Review and Discussion on Real-Time Lightning 3D Imaging and Forecasting Project for Sustainable and Reliable Energy Supply of Energy and Storm Disaster Early Warning

UTeM
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MINISTRY OF HIGHER EDUCATION

Project Overview

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Real-Time Lightning 3D Imaging and Forecasting Project for Sustainable and Reliable Supply of Energy and Storm Disaster Early Warning

For the lightning activity in the Straits of Malacca, 3D lightning observation network is realized for where the discharge initiates, how it progresses, and where it terminates. Real-time estimation of the charge distribution in the thundercloud and nowcasting information is provided to relevant institutions.

They contribute to the stable electric power supply and application to early warning for severe weather disasters.

SATREPS For the Earth, For the Next Generation

Science and Technology Research Partnership for Sustainable Development

The SATREPS program is promoting international joint research with the goal of resolving global issues as one aspect of "science and technology diplomacy," which connects science and technology with diplomacy for the mutual advancement of both.

In the SATREPS program, researchers from Japan and partner countries are together tackling issues in the four fields of Environment and Energy, Bioresources, Disaster Prevention and Mitigation, and Infectious Disease Control, discovering new knowledge and technologies with real-world applications in light of local needs, and thus contributing to the international community that is working toward sustainable development.

Research contents and plan

Subject 1

High-resolution observation of lightning channel development by EM measurement in various frequencies

What kind of lightning?

Subject 2

Real-time 3D imaging/estimation of charge distribution inside thunderclouds

What kind of thundercloud?

Subject 3

Study on short-time forecasting of CG lightning and the correlation between lightning activity and heavy rain

Contribute to sustainable energy supply and alert for sever weather disasters

What will happen in near future?

How can we use obtained data?

Output 1.

The lightning observation network obtaining the data set for real-time 3D imaging is developed

1-1 Install & observe lightning discharge by LF network: > 9 stations
Imaging lightning channel, coverage : > 100 km

1-2 Install & observe lightning discharge by VHF network: > 6 stations
Imaging lightning leader development, coverage : tens kilometers

1-3 Install & observe high-energy radiation : 3 Gamma-ray & Radon

1-4 high-speed video camera

1-5 Install & conduct E-field measurement : 5 sets

1-6 Develop the dataset for the model of real-time 3D imaging

Realize lightning channel imager with sufficient resolutions
for real-time 3D estimation of charge distribution
(100 m resolution in 1 min.)

Output 1.

The lightning observation network obtaining the data set for real-time 3D imaging is developed

Detailed Design for the equipment and procurement

1-1 LF systems : 9 sets

Recorders (Software Radio device)	Handed over
Sensors (2 systems with Slow antenna)	Purchase ordered
PCs	Model selection

1-2 VHF systems: 6 sets — Purchase ordered

1-3 Gamma-ray & Radon monitors : 3 sets — Detailed designed

1-4 High-speed video camera : 2 sets — [MY] Procurement process

1-5 E-field measurement : 5 sets

Sensors	Handed over
Data loggers	Consideration in UNITEN

Output 1.

The lightning observation network obtaining the data set for real-time 3D imaging is developed

Site selection and background EM noise survey

Site	Use for	Responsible	Remarks
DID, Batu Dam	LF, VHF	UNITEN	
DID, Padang Jawa	LF, VHF	UNITEN	
UNITEN, Putrajaya Campus	LF, VHF	UNITEN	
MMD, Kuala Pilar	LF	UNITEN	
Kolej UNITI, Port Dickson	LF	UTeM	
Falak Astronomy Complex, Malacca	LF	UTeM	
UTeM, Malacca	LF, VHF, RTL	UTeM	VHF pilot obs.
UiTM Jasin Campus	LF, VHF, RTL	UTeM	
Pulau Besar	LF, VHF	UTeM	

Output 2.
Real-time lightning 3D imaging is composed and evaluated using lightning observation network

2-1 Develop real-time lightning 3D imaging model
for charge location and charge quantity

2-2 Conduct lightning current measurement : 7 sets for 2 types of structure

2-3 Conduct rocket triggered lightning experiment
Japan (2023～), Malaysia (2025～)

in order to validate the real-time lightning 3D imaging model

2-4 Improve the real-time lightning 3D imaging model for nowcasting
based on the 2-2 and 2-3.

3D charge distribution : 500 m and 0.01 C resolution in 5 min.
Lightning current waveforms : 20 events/year
Rocket triggered lightning : 40 events (total)

Output 2.
<u>Real-time lightning 3D imaging is composed and evaluated using lightning observation network</u>

2-1 Develop real-time lightning 3D imaging model

Examined and trialed with past data

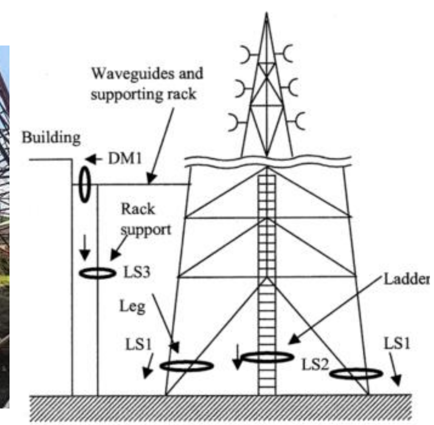
2-2 Conduct lightning current measurement : 7 sets for 2 types of structure

Tall building : KL tower → pending

Communication Tower : Bukit Gasing Transmission Tower

Proceed with the measurement by 5 sensors first.



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2-3 Conduct rocket triggered lightning experiment

Japan (2023～), Malaysia (2025～)

RTL Training in Kanazawa with 12 MY researchers.

Lectures to make the related understanding deepened.

2 MY RTL sites are decided.

Output 3.
Social implementation products of nowcasting, alert system, and countermeasure for lightning damage are developed

3-1 Conduct pilot operation of nowcasting for lightning activity.

3-2 Develop and test lightning alert system in study area

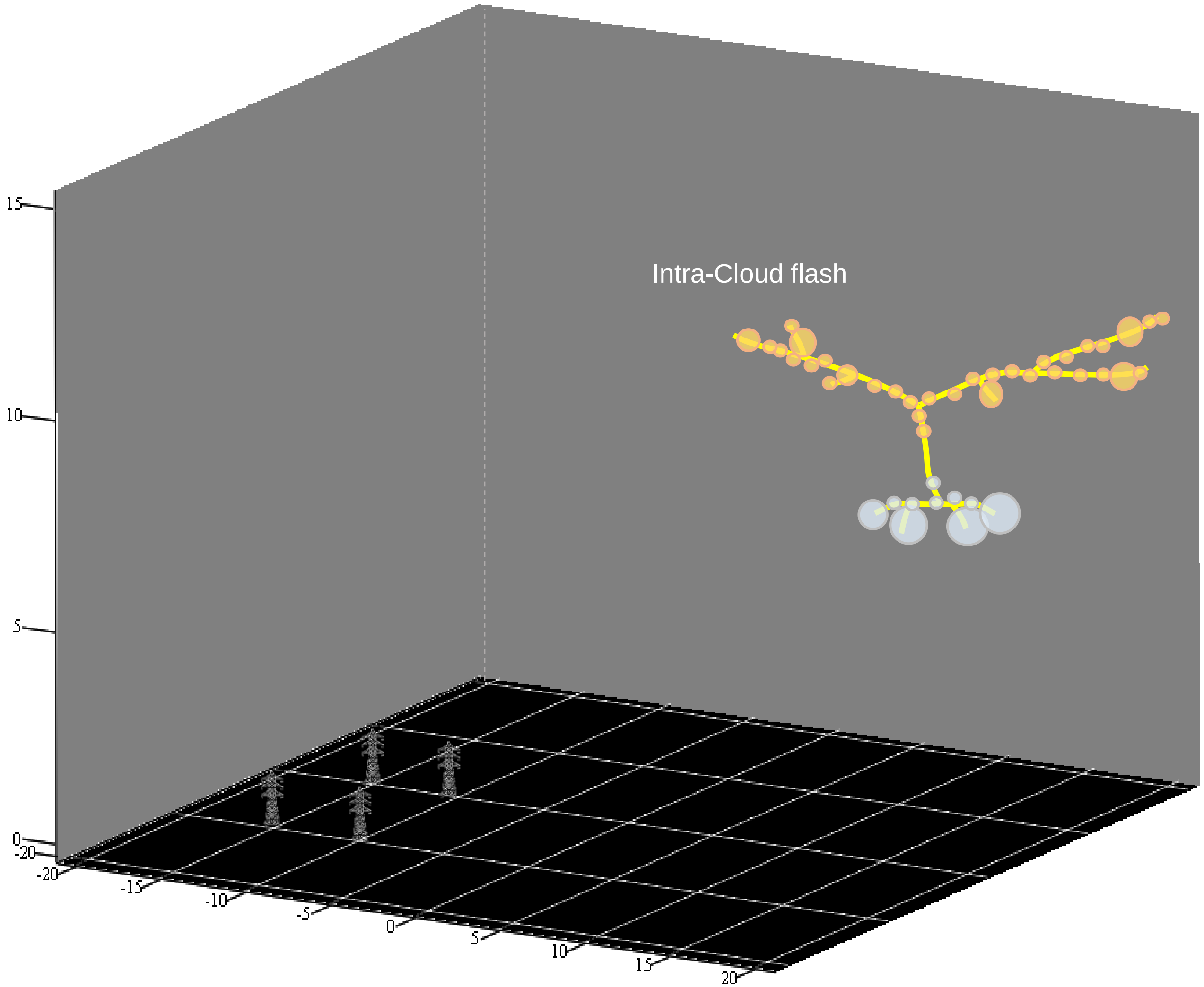
3-3 Develop the countermeasure for lightning damage using IoT technique

3-4 Enhance the opportunities for using the project output through workshop

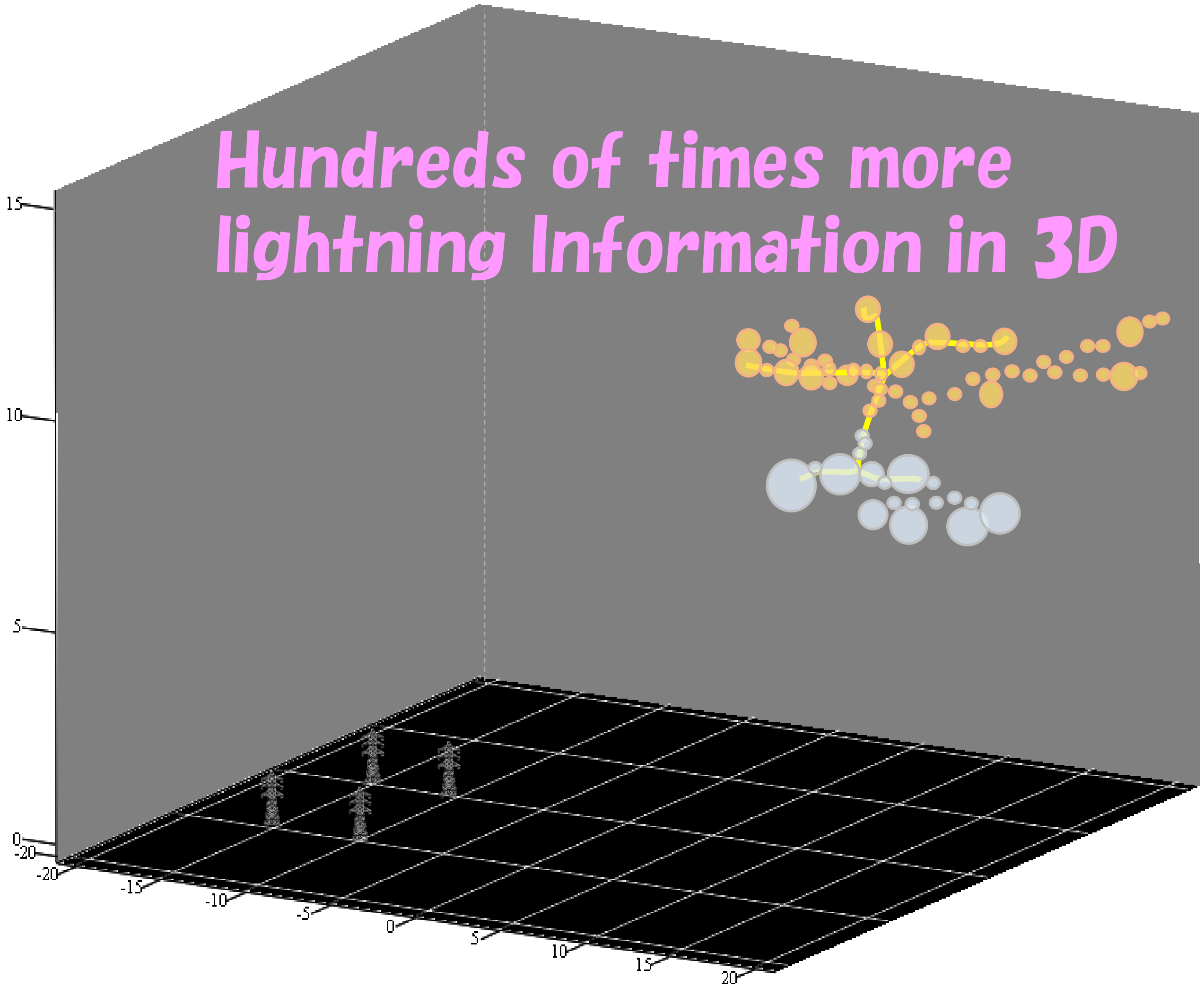
3-5 Propose social implementation products to potential users.

CG lightning forecasting : Hit rate 50%, 5 min. lead time
Heavy rain (>50mm/h) prediction before 15 min.

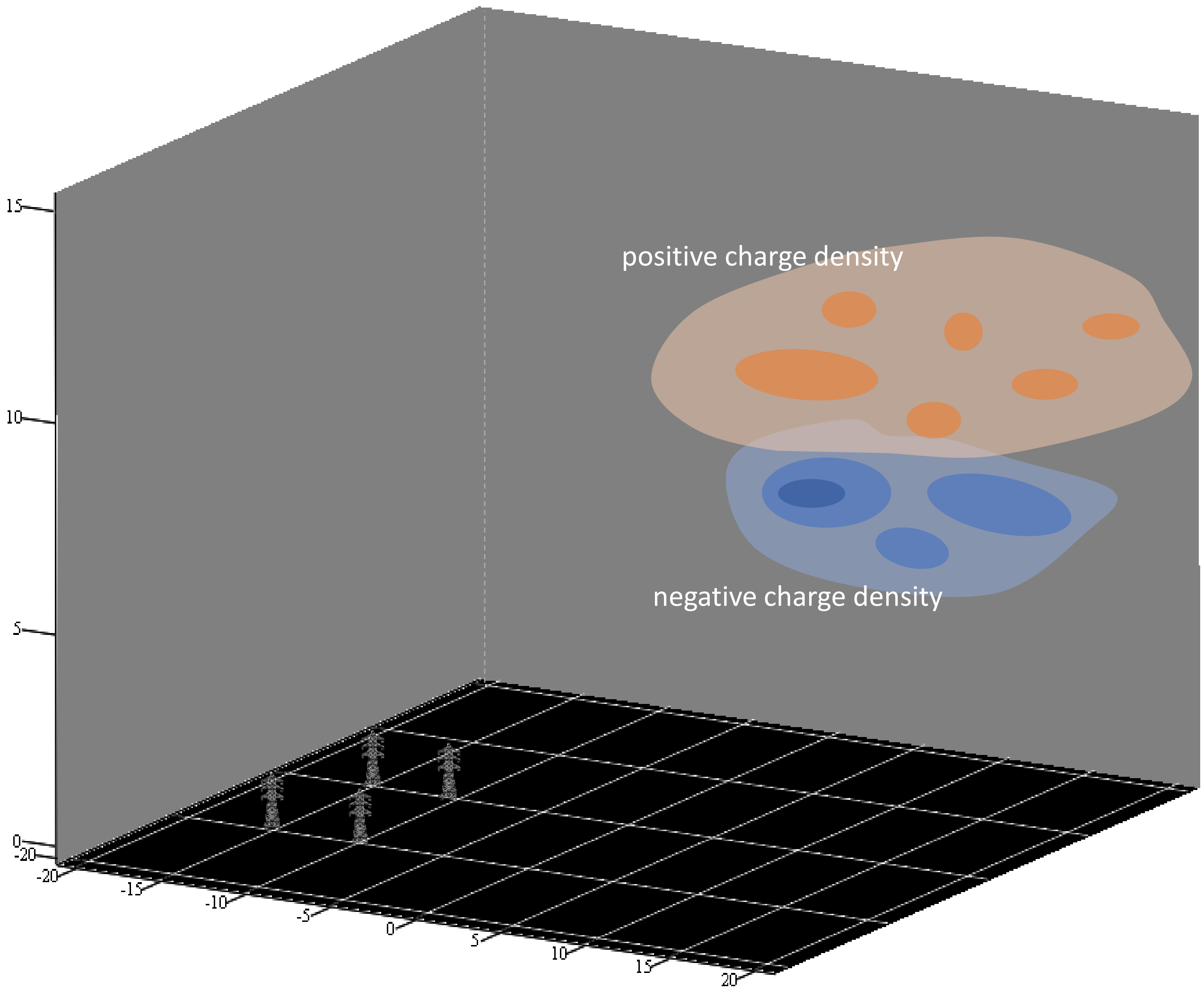
3D estimation of charge distribution



3D estimation of charge distribution



Toward short-time forecasting for CG lightning



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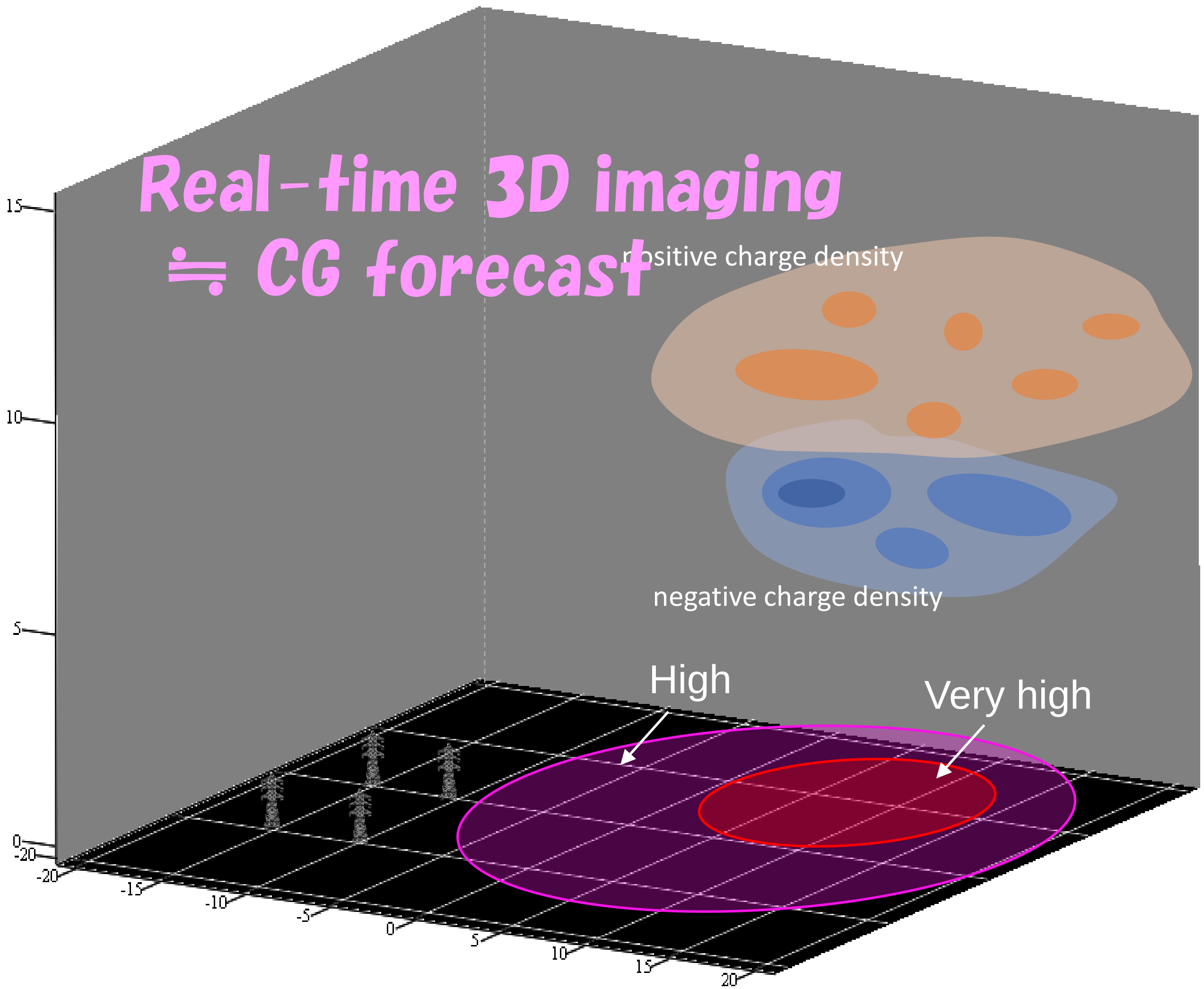

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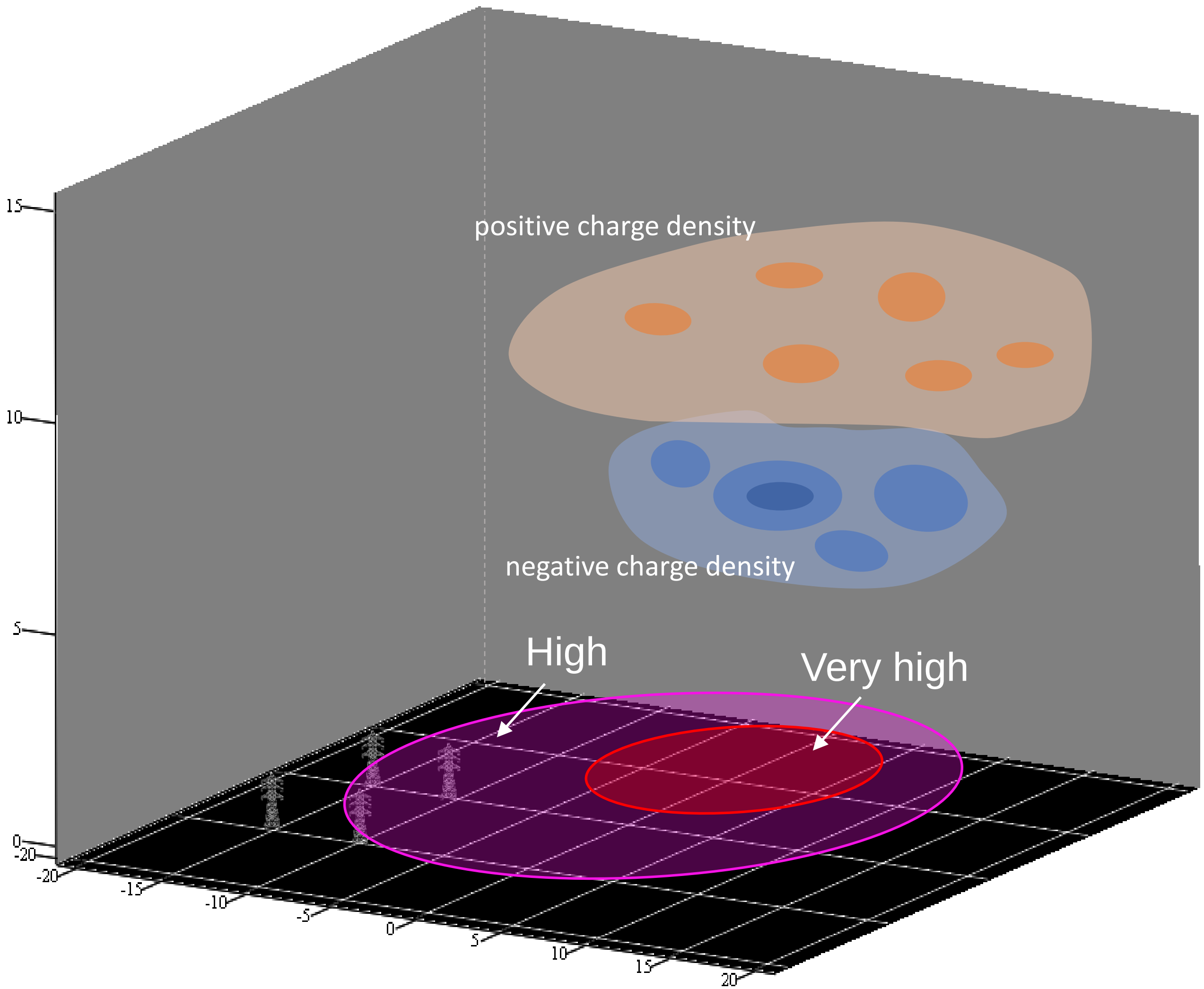


Output 3.
Social implementation products of nowcasting, alert system, and countermeasure for lightning damage are developed

3-1 Conduct pilot operation of nowcasting for lightning activity.

3-2 Develop and test lightning alert system in study area

DID and MMD became Joint Research Institution

3-3 Develop the countermeasure for lightning damage using IoT

Switching unit : Prototype 1 was developed

IoT unit : Joint development

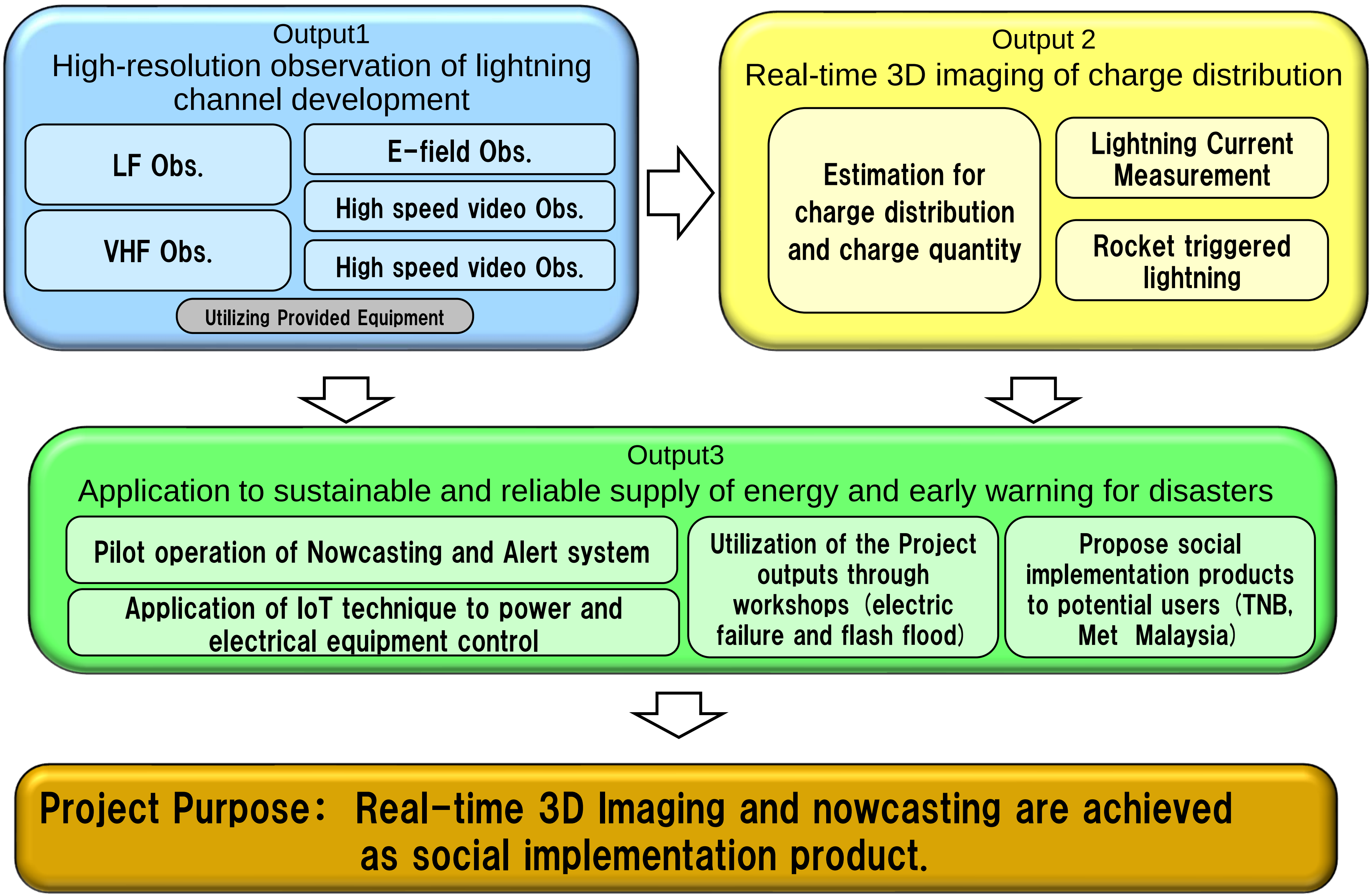
3-4 Enhance the opportunities for using the project output through workshop

3-5 Propose social implementation products to potential users.

Seminar, Scientific conference, Workshop

Demonstration with real obtained data is expected

Real-Time Lightning 3D Imaging and Forecasting Project for Sustainable Energy Supply and Storm Disaster Early Warning：Master Plan Image



Output 1.
The lightning observation network obtaining the data set for real-time 3D imaging is developed

1-1 Install & observe lightning discharge by LF network
: 6 → 9 stations

Imaging lightning channel

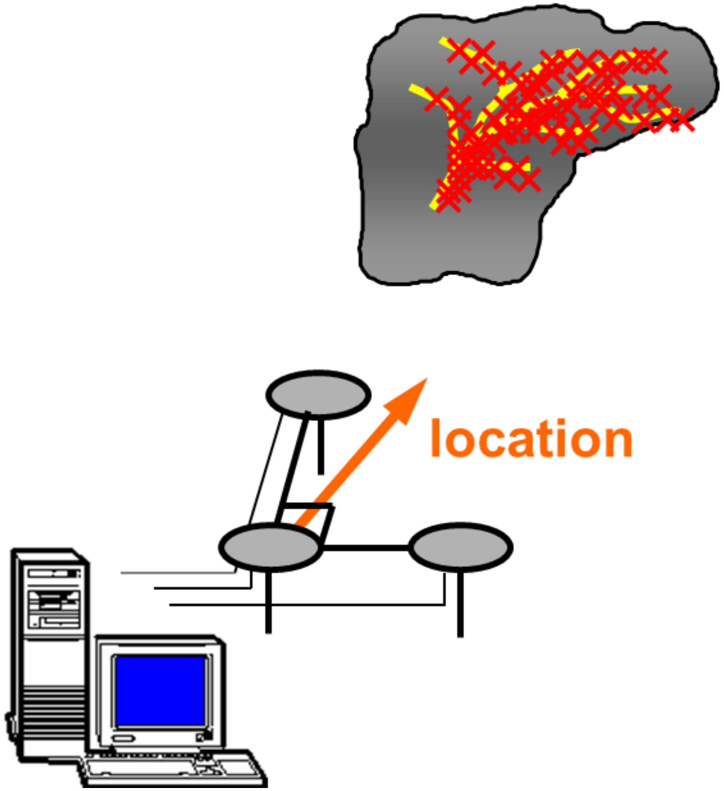
Coverage : more than hundred kilometers

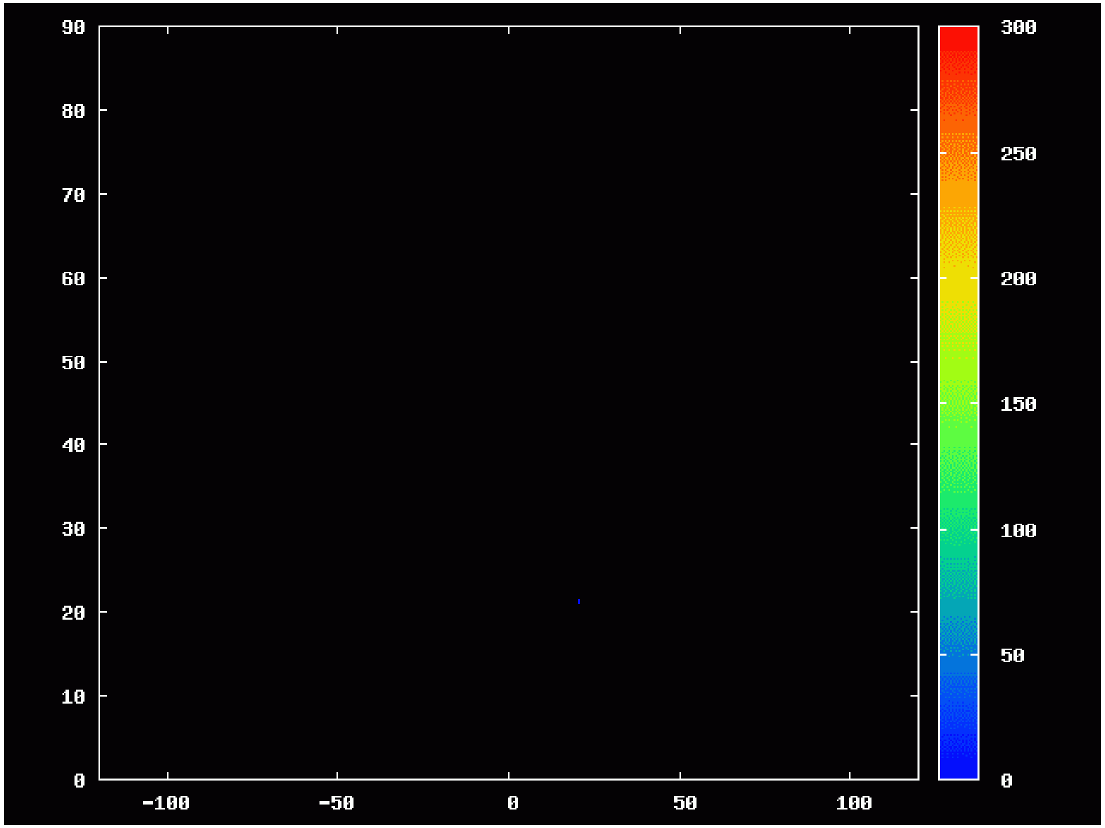
1-2 Install & observe lightning discharge by VHF network
: 6 stations

Imaging lightning leader development

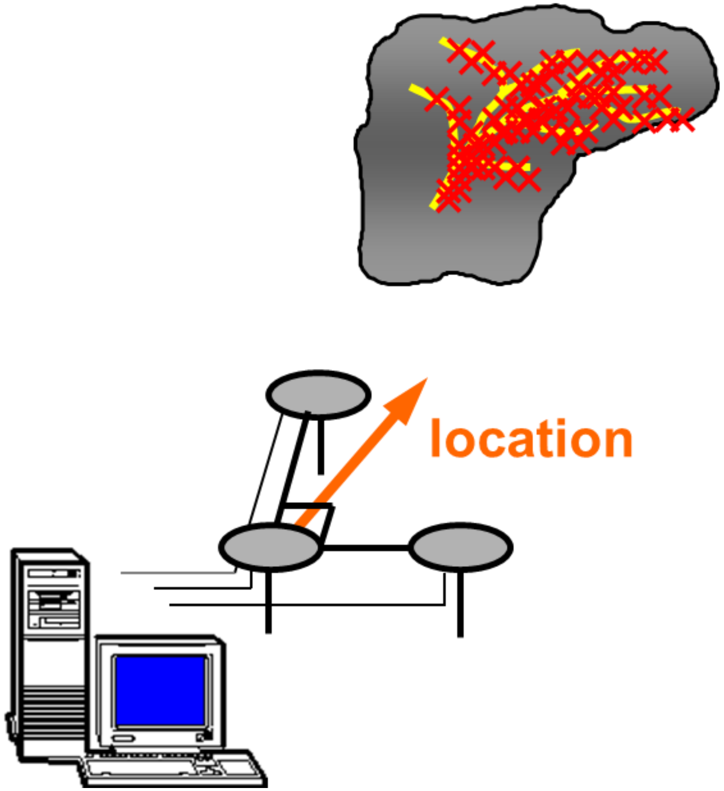
Coverage : several tens of kilometers

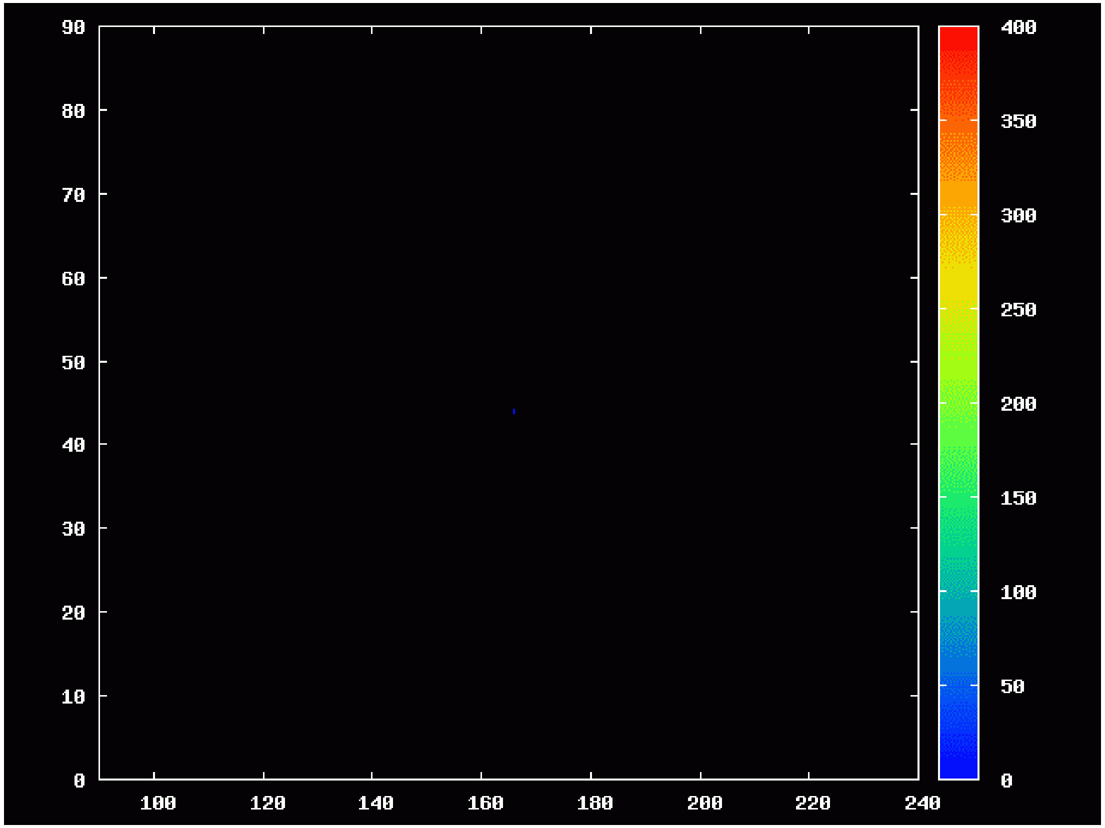
VHF System (Lightning Channel Imager)



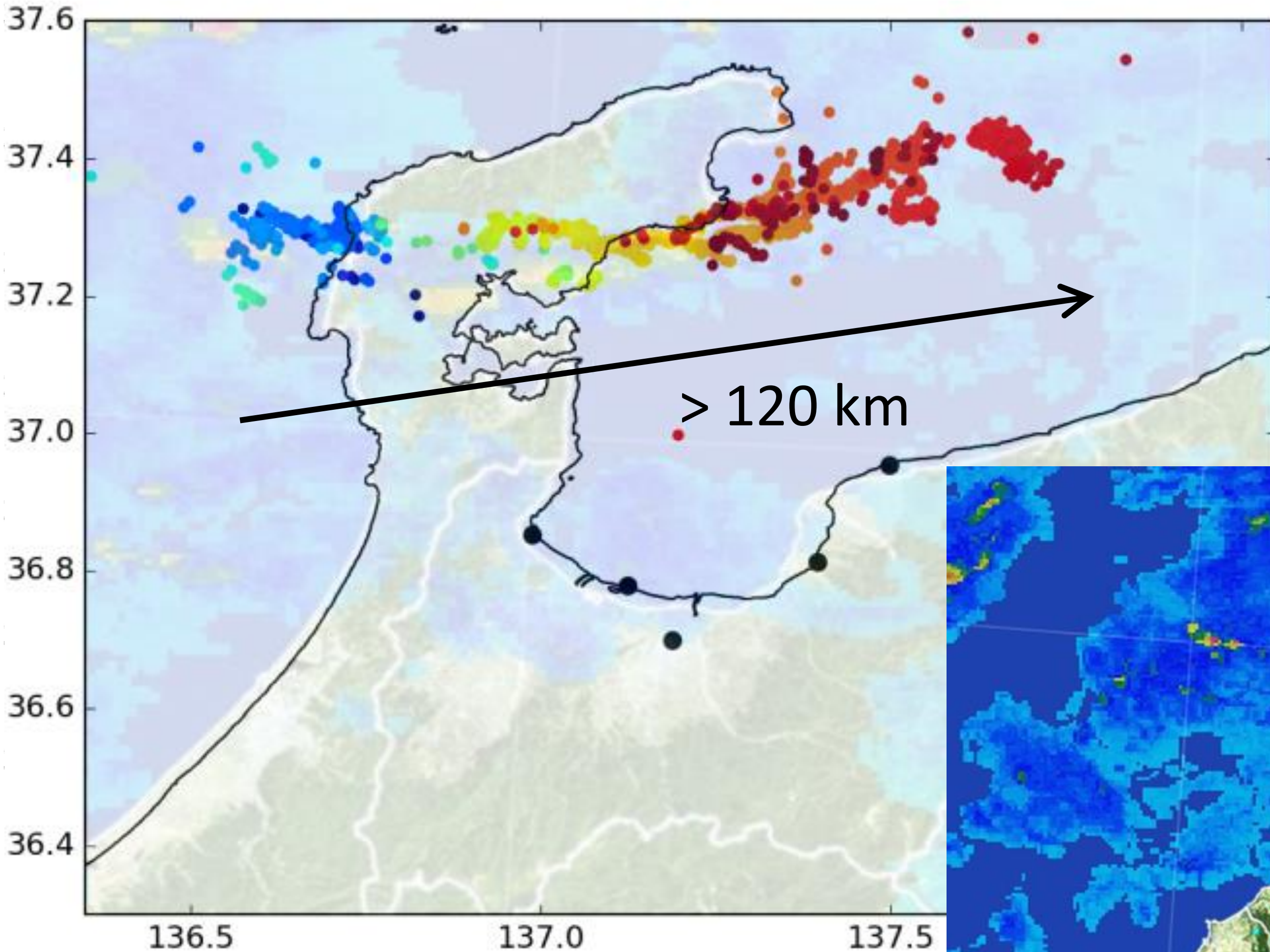


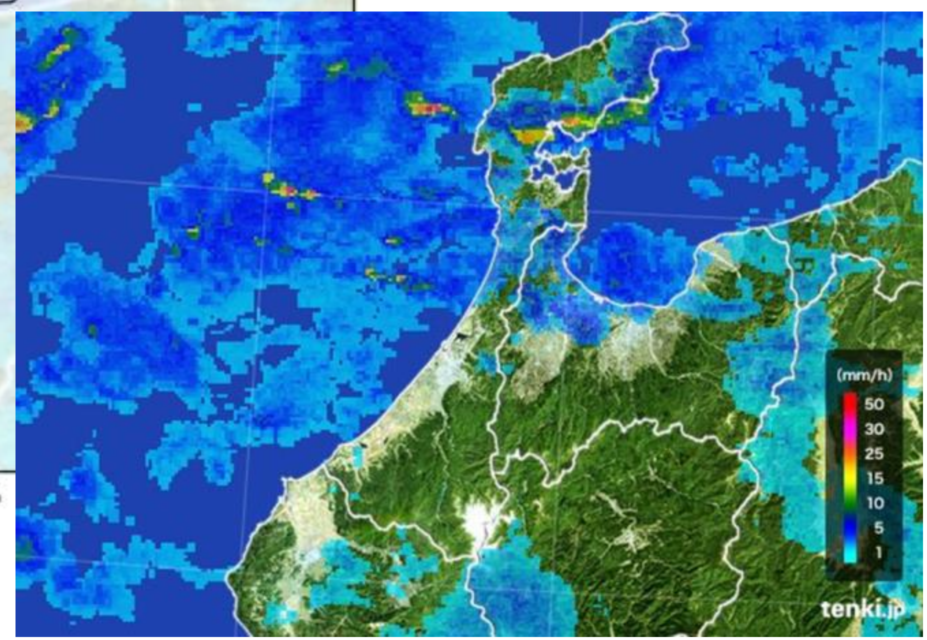
VHF System (Lightning Channel Imager)



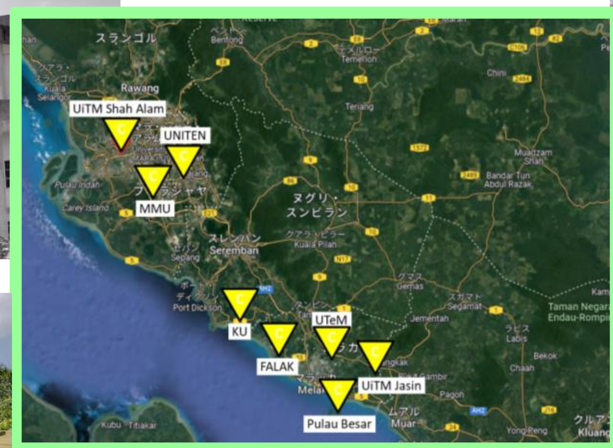


LF wide-area lightning observation network





EM noise survey



Output 1.

The lightning observation network obtaining the data set for real-time 3D imaging is developed

1-1 Install & observe lightning discharge by LF network: > 7 stations

Imaging lightning channel, coverage : > 100 km

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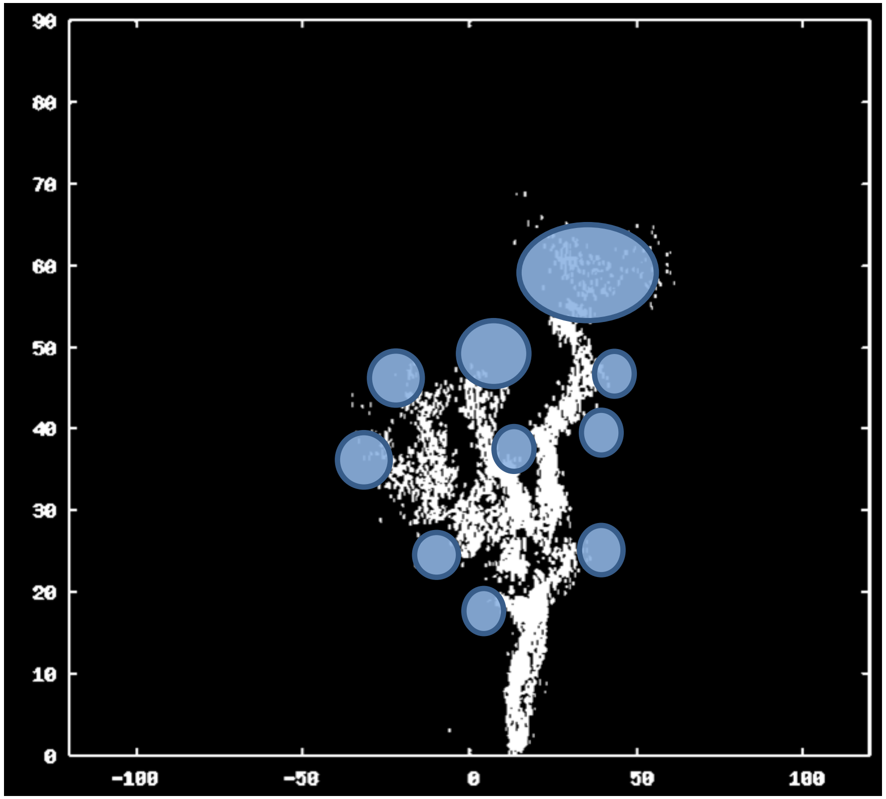
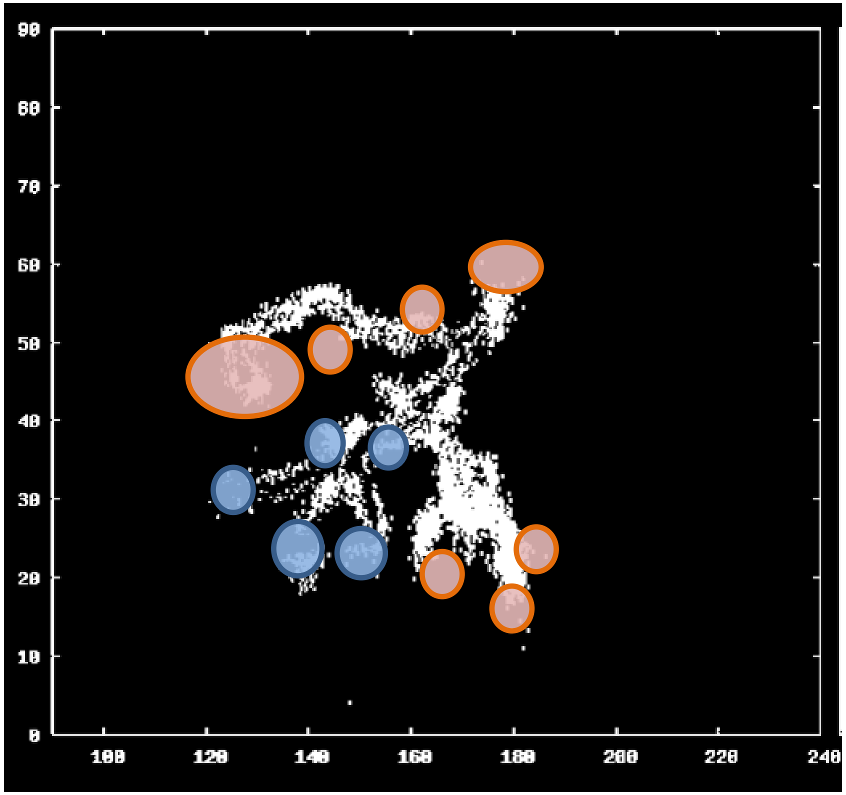
Output 2.
Real-time lightning 3D imaging is composed and evaluated using lightning observation network

2-1 Develop real-time lightning 3D imaging model
for charge location and charge quantity



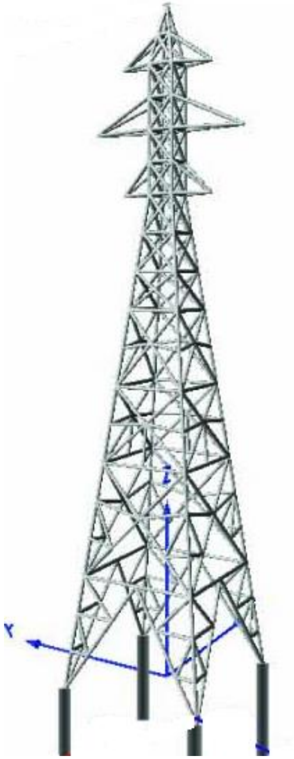


Output 2.
Real-time lightning 3D imaging is composed and evaluated using lightning observation network

2-2 Conduct lightning current measurement : 5 sets

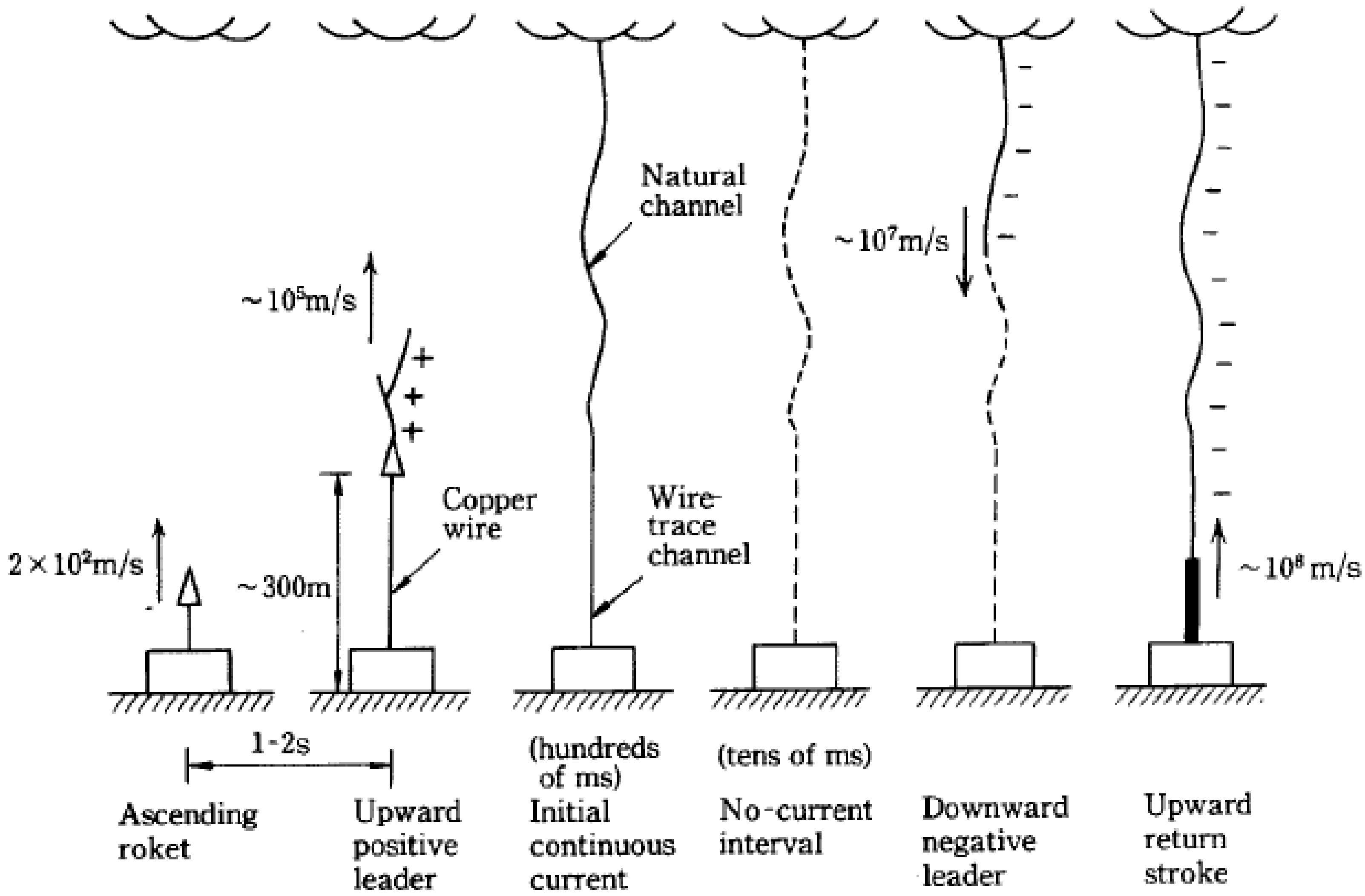
2-3 Conduct rocket triggered lightning experiment
Japan (2023～), Malaysia (2025～)

in order to validate
the real-time lightning 3D imaging model



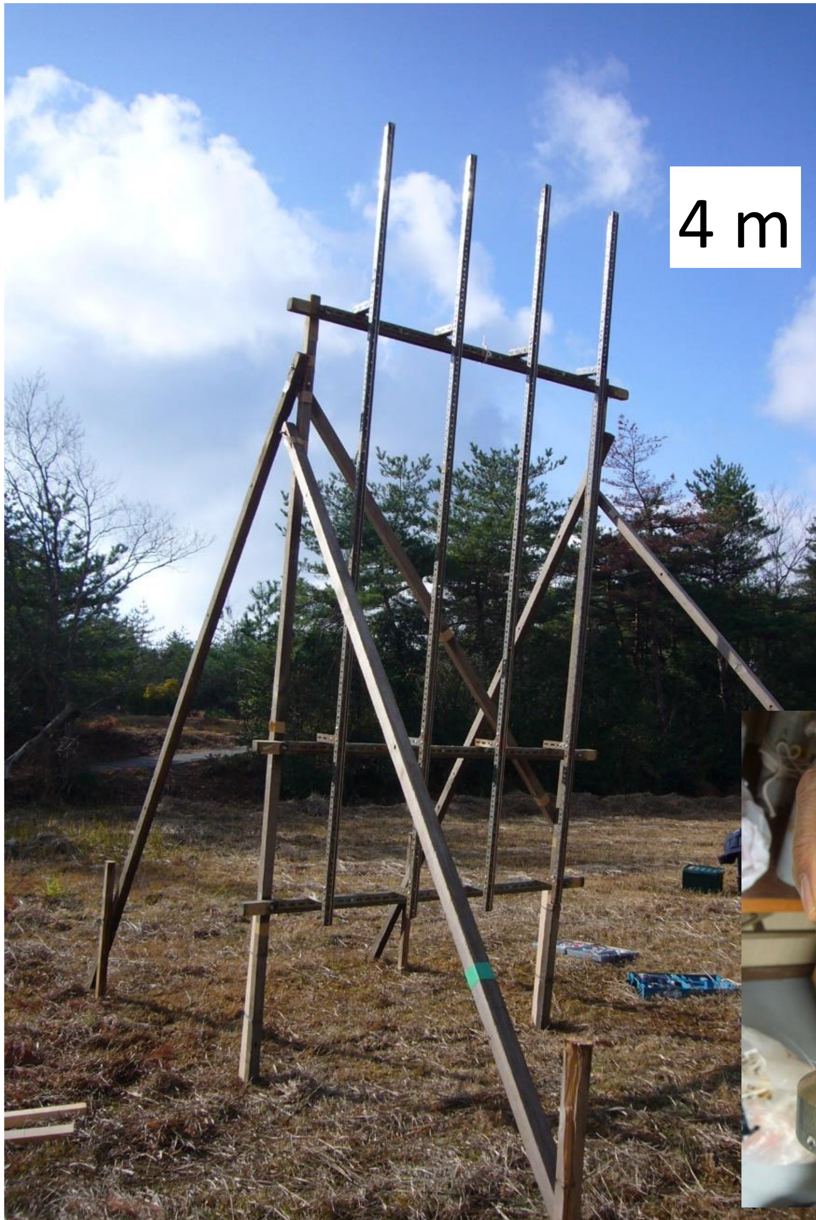
Rocket Triggered Lightning Experiment

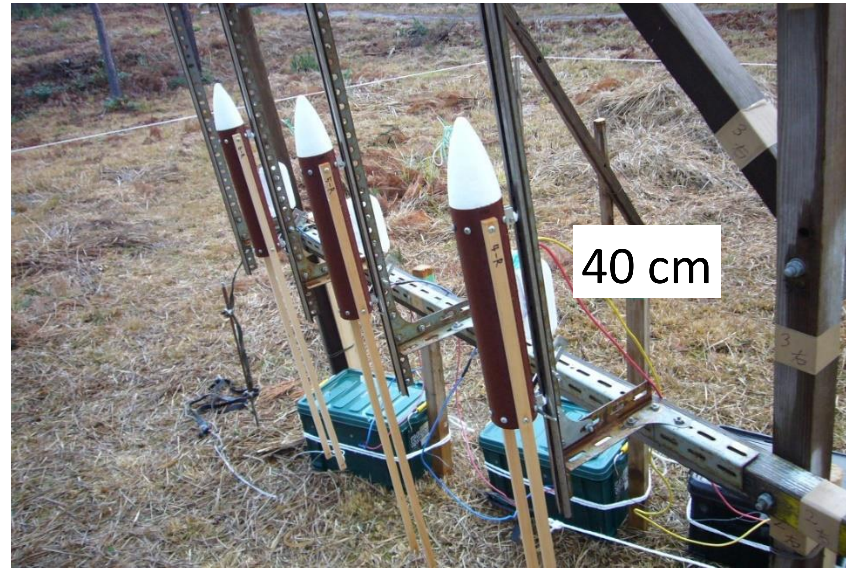
Rocket Triggered Lightning Experiment

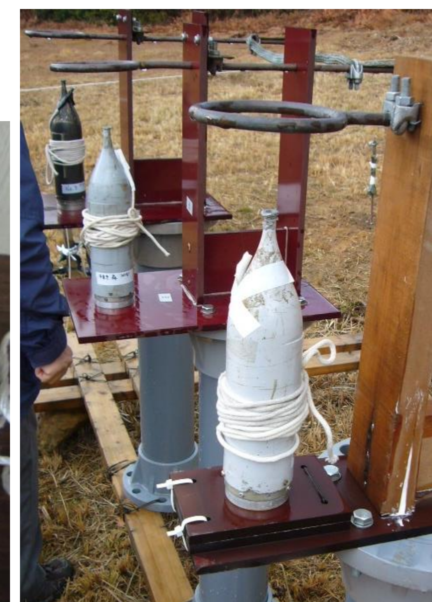


M.A.Uman et al. (1997)

Rocket Triggered Lightning Experiment







Output 3.
Social implementation products of nowcasting, alert system, and countermeasure for lightning damage are developed

- The 3D leader channels and the charge distribution inside cloud can be regarded as immediate predictive information for upcoming lightning and thunderstorm activity

- Significant amount of research to study the relationship between lightning and rainfall

- Research achievements related to flood forecasting

Research contents and plan

Outputs and activities \ Year	2022 (preparation)	2023	2024	2025	2026	2027
1. The lightning observation network obtaining the data set for real-time lightning 3D imaging is developed.	Site survey & arrangement	Observation	at the installed	Site	Continuous	observation
1-1. Install and observe lightning discharges by LF network.		Equipment preparation	Installation			
1-2. Install and observe lightning discharges by VHF network.						
1-3. Install and observe high-energy radiation.						
1-4. Install and observe high-speed video camera.						
1-5. Install and conduct E-field measurement.						
1-6. Develop the data set for developing the model for real time 3D imaging.			Consideration		Operation &	delivery
2. Real-time lightning 3D imaging model is composed and evaluated using the lightning observation network.	Study for	data in Japan	Observation,	estimation,	verification,	adjustment
2-1. Develop real-time lightning 3D imaging model for charge location and charge quantity.			Consideration		Operation	
2-2. Conduct lightning current measurement in order to validate the real-time lightning 3D imaging model.	Site suevey& arrangement					
2-3. Conduct rocket triggered lightning experiment in order to validate the real-time lightning 3D imaging model.		Experiment	site preparation		Field experiment	
2-4. Improve the real-time lightning 3D imaging model for nowcasting based on the 2-2 and 2-3.						
3. Social implementation products of nowcasting, alert system, and countermeasure for lightning damage are developed.	Search for	past cases				
3-1. Conduct pilot operation of nowcasting for lightning activity.				Consideration	implementation	
3-2. Develop and test lightning alert system in study area.						
3-3. Develop the countermeasure for lightning damage using IoT technique.				Consideration	implementation	protptype
3-4. Enhance the opportunities for using the Project outputs through workshops.						
3-5. Propose social implementation products to potential users.						

Research contents and plan

Year / Outputs and activities	2022 (preparation)	2023	2024	2025	2026	2027
1. The lightning observation network obtaining the data set for real-time lightning 3D imaging is developed.	Site survey & arrangement	Observation	at the installed	Site	Continuous	observation
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Overall Goal:
Real-time lightning 3D imaging and nowcasting system is utilized to reduce the lightning damages and electric failures.

- Real-time Lightning 3D Imaging and nowcasting are utilized by Malaysian organizations and contribute to disaster risk reduction for safety of public people, transportation operation (aviation, maritime, etc.) and industry (energy, fishery, tourism, construction, ICT, etc.)
- Real-time Lightning 3D Imaging and nowcasting are expanded in other areas of Malaysia and neighboring countries.
- Real-time Lightning 3D Imaging and nowcasting contribute to forecasting or alert information for severe disaster such as heavy rain and flash flood through comparing thunderstorm activity with other information such as meteorological data, GIS (Geographical Information System) and etc.

Main Research Members

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D. Wang
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M. Akita
(Univ. of
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Z. Kawasaki
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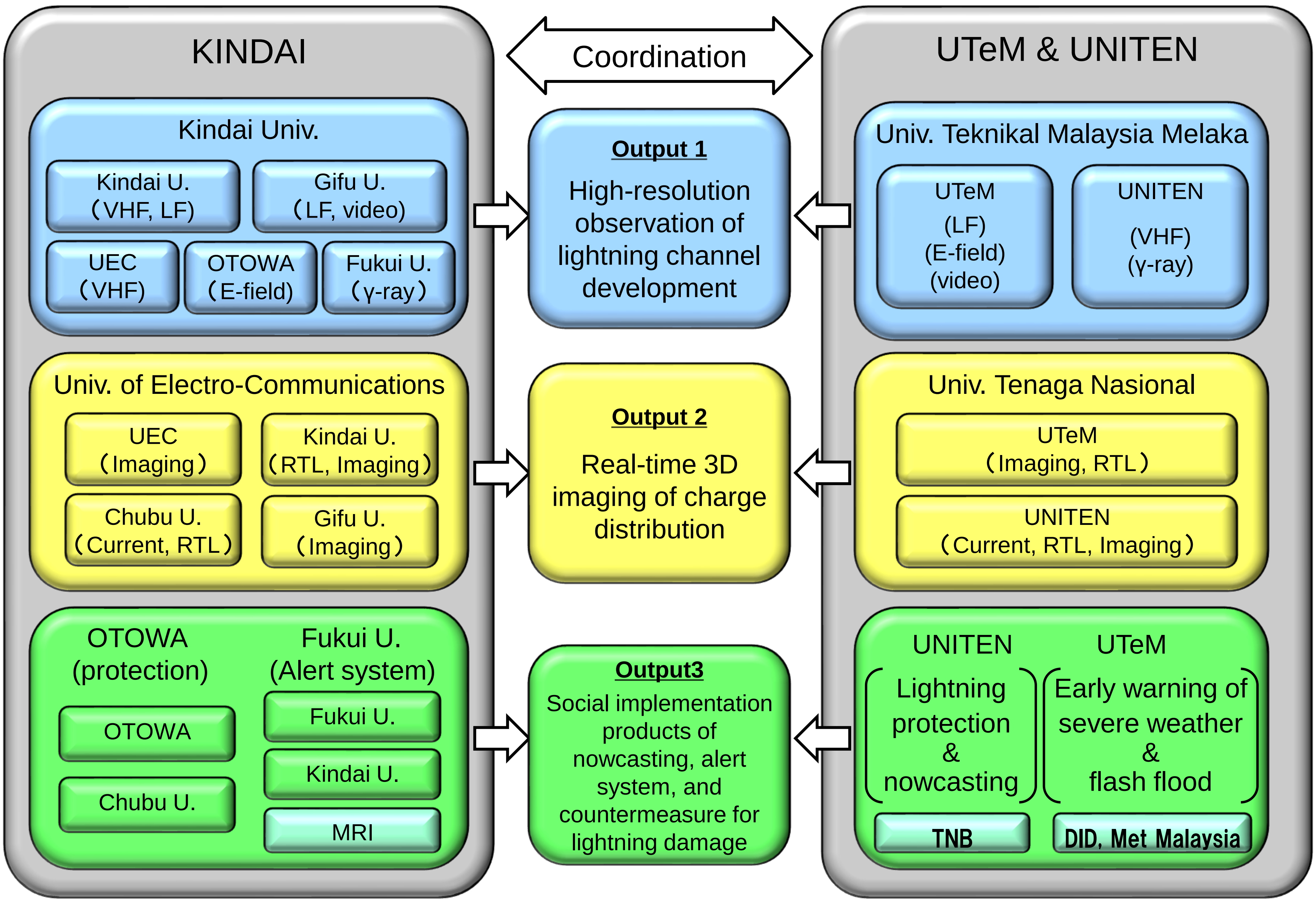
Osaka Univ.)

Real-Time Lightning 3D Imaging and Forecasting Project for Sustainable Energy Supply and Storm Disaster Early Warning：Operation Image



We sincerely welcome your support, cooperation, suggestions, questions, requests etc..

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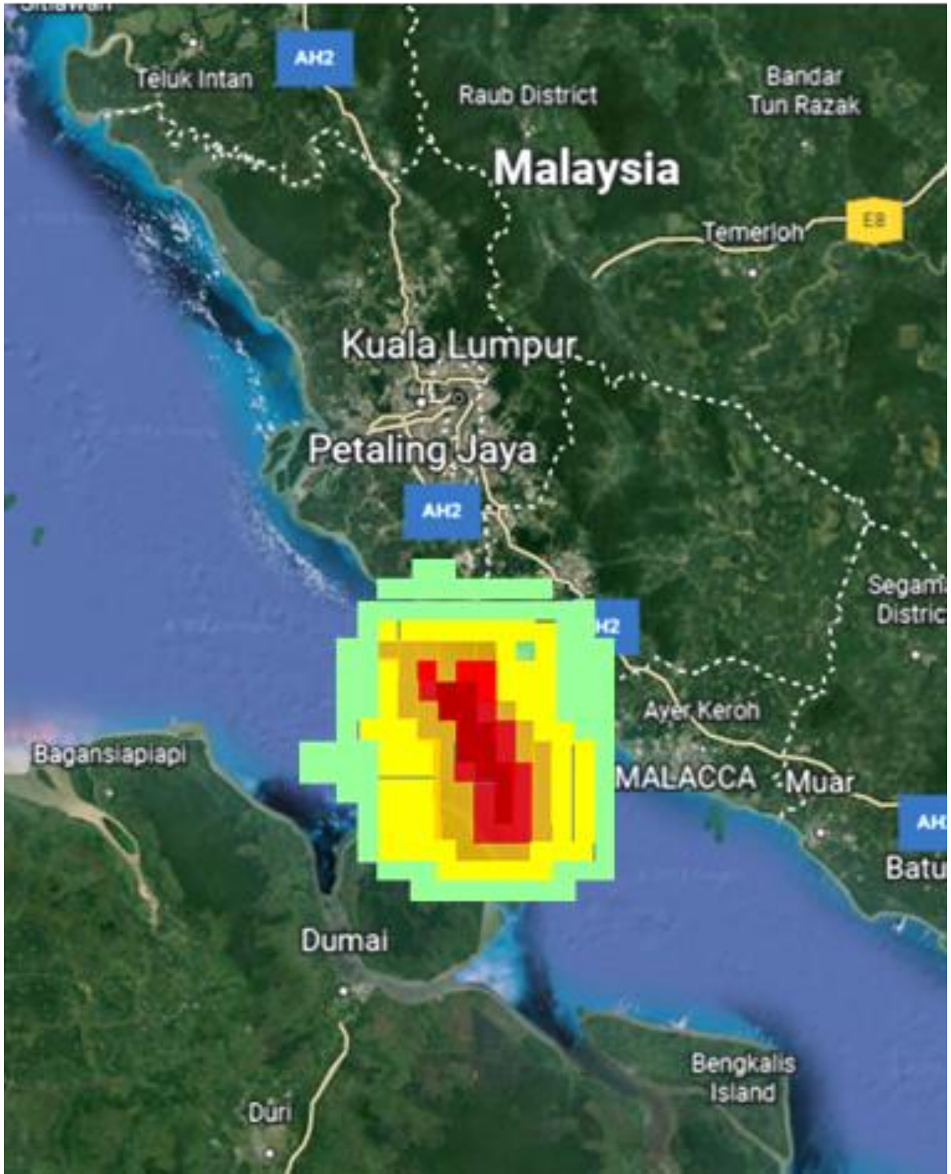
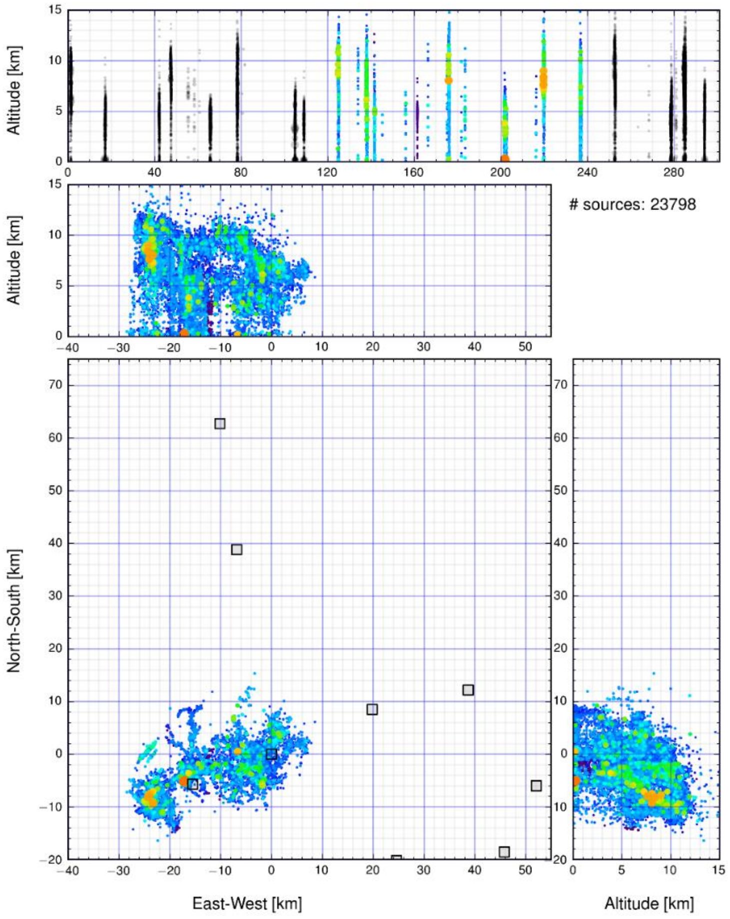
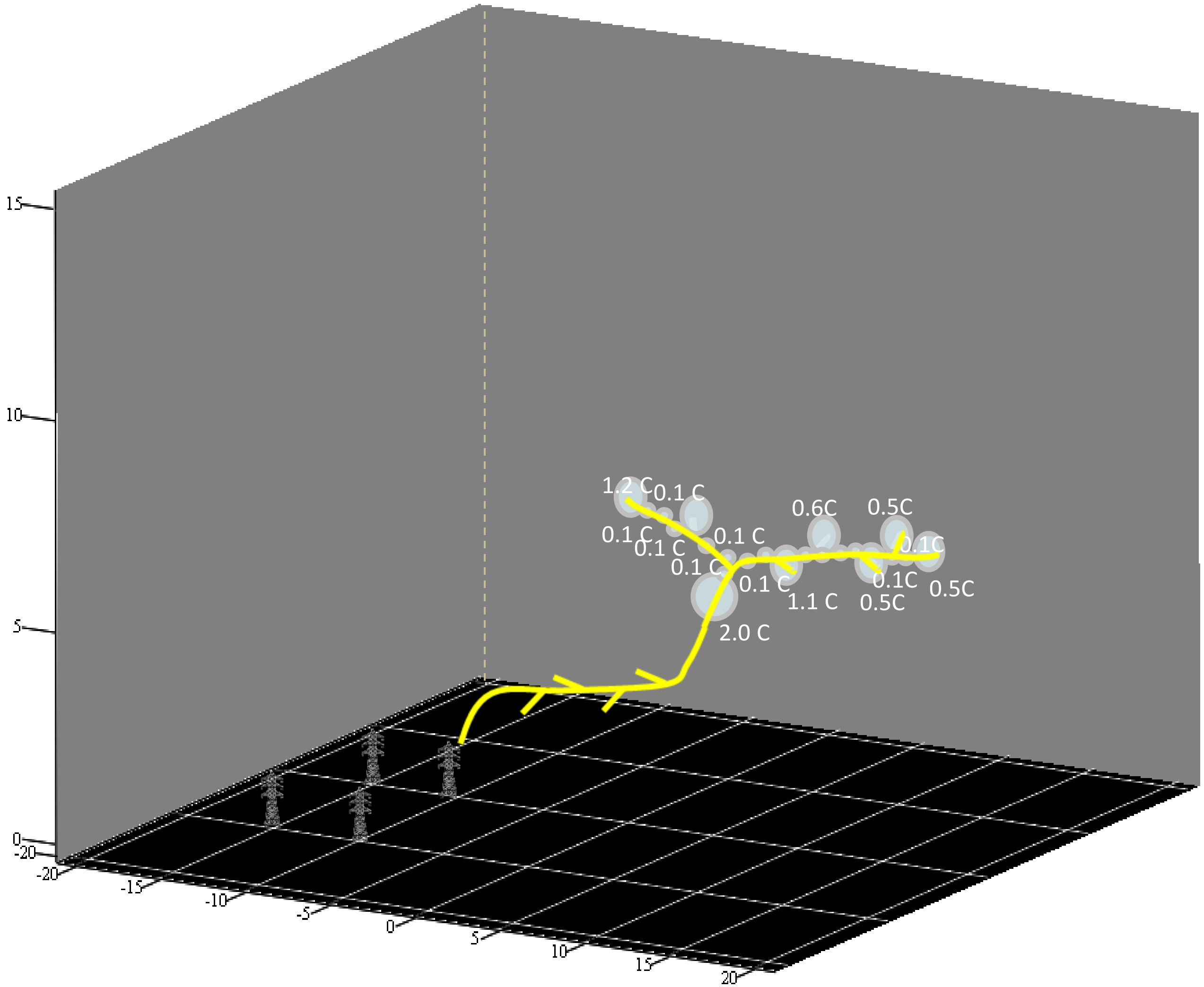
UiTM Shah Alam
UNITEN
MMU
KU
FALAK
UTeM
UiTM Jasin
Pulau Besar

Integrated 3D charge distribution

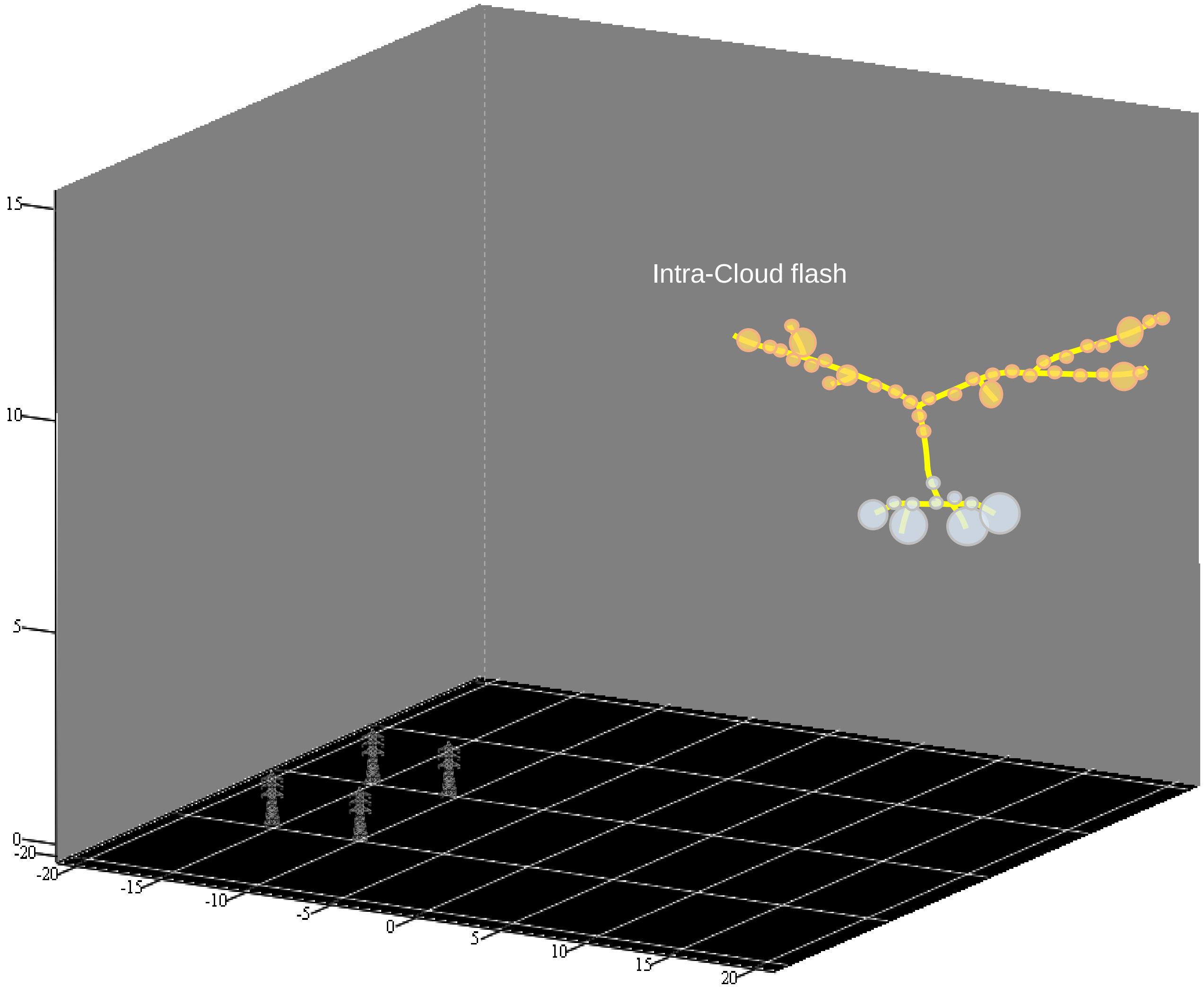




3D estimation of charge distribution



3D estimation of charge distribution



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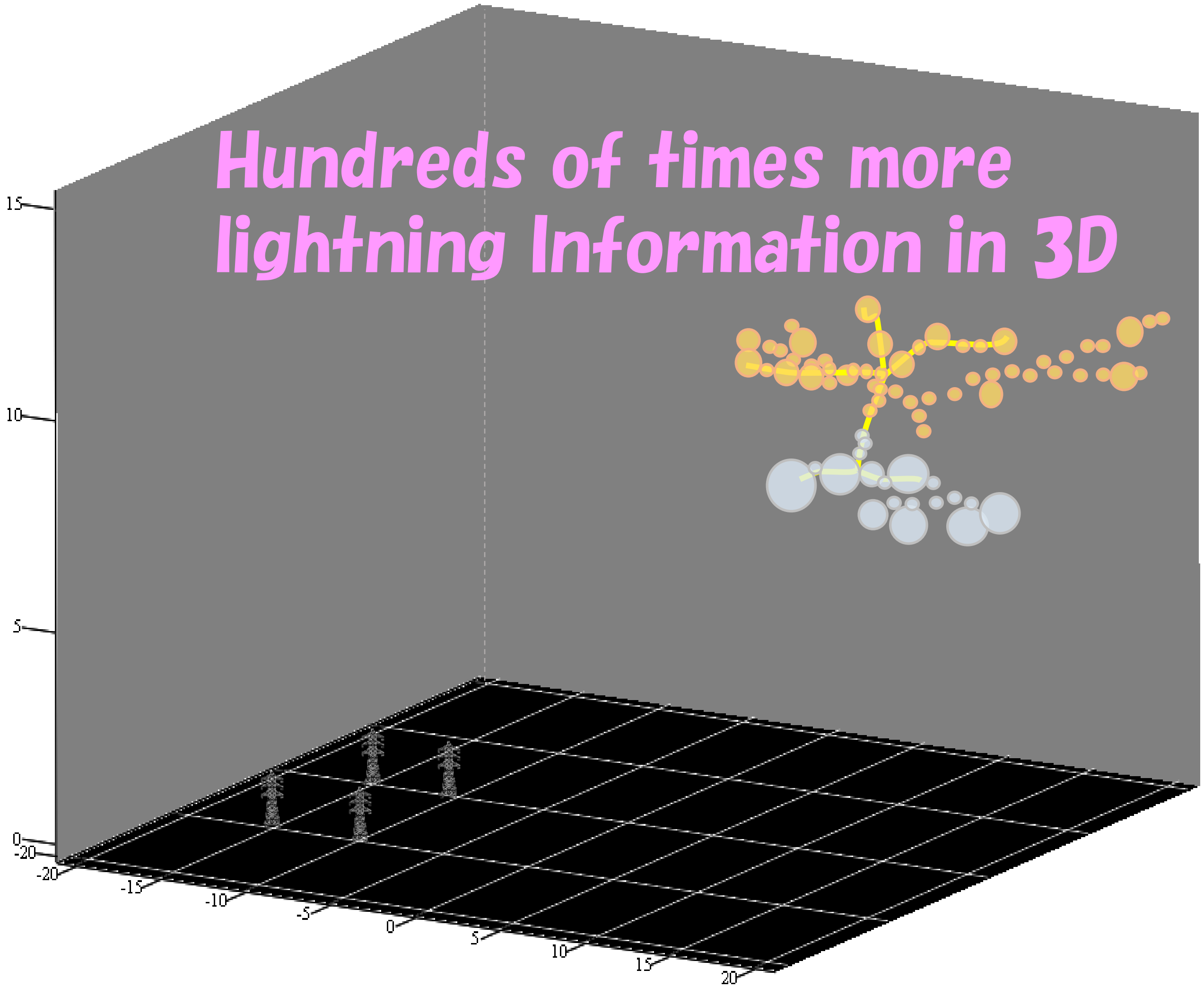
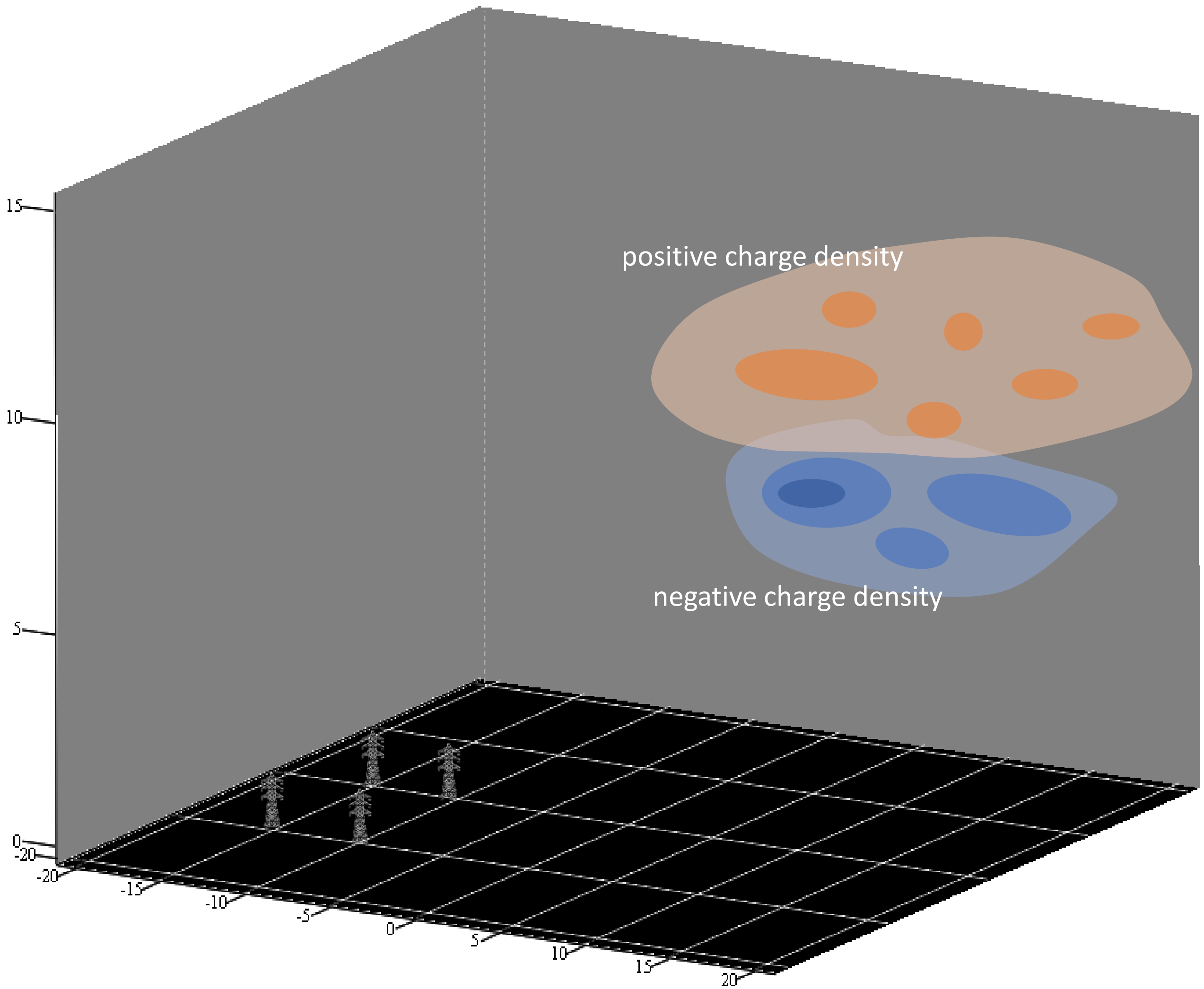


Toward short-time forecasting for CG lightning



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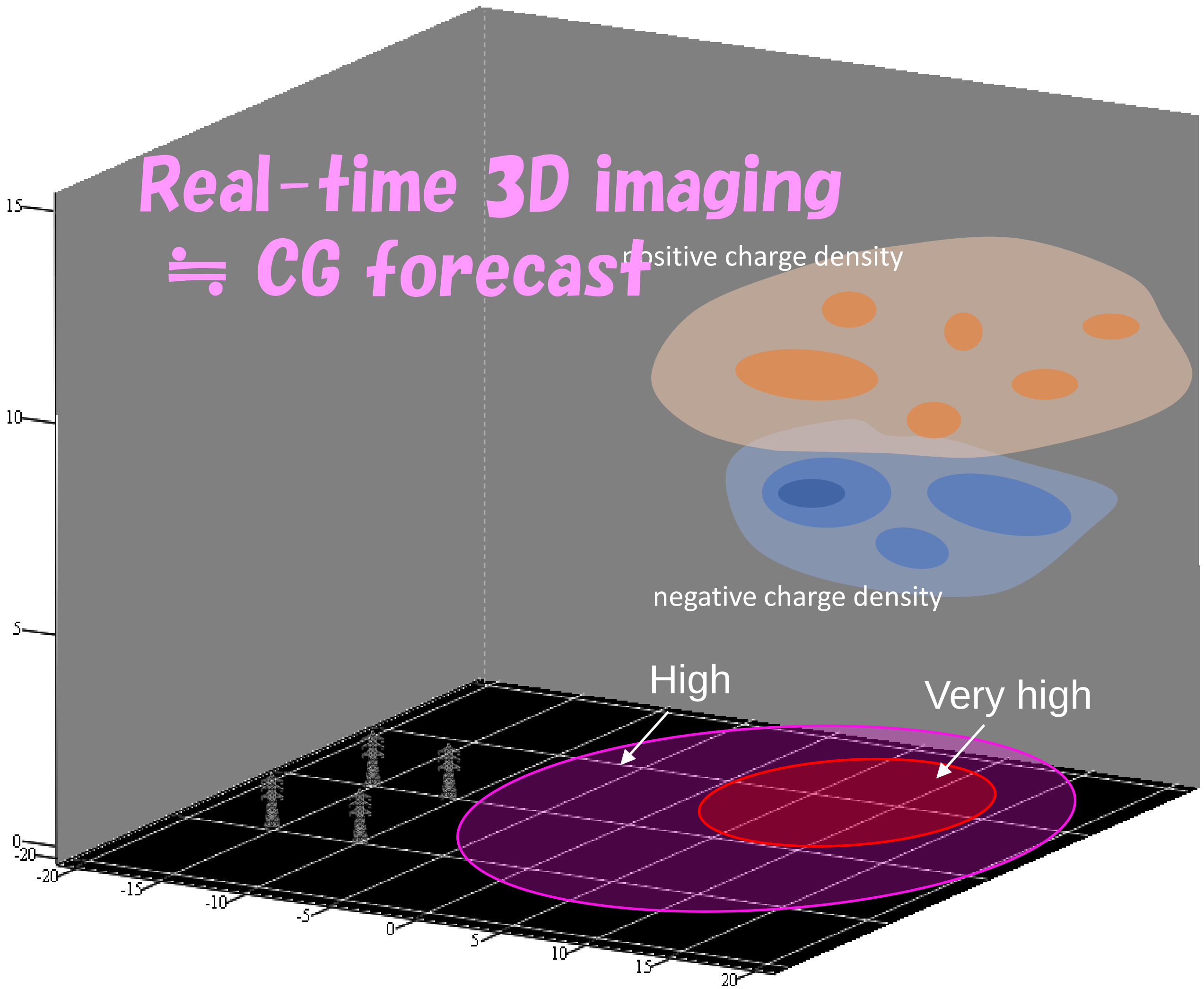

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